Lisa Nguyen

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Education

University of Michigan **Expected Graduation: Dec 2026**
PhD in Applied Physics and Scientific Computing
Topic: Spatio-Temporal Dynamics and Causal Drivers of Hydro-climatic Extremes Under Climate Change

University of California Berkeley **Graduation: Dec 2019**
B.A. in Physics; B.A. in Pure Mathematics

Awards/Fellowships/Programs

- 2026 AGU Travel Grant
- Rackham Predoctoral Fellowship, 2026-27 academic year (≈85 selected campus-wide)
- University of Michigan student delegate for COP30 (1 of 8 candidates selected campus-wide)
- Thriving Earth Exchange Community Science Fellowship
- Future of Research in Climate, Earth, and Energy (FORCEE) DoE Internship
- Rackham Merit Fellowship
- Joint PhD program in Scientific Computing, Michigan Institute for Computational Discovery and Engineering (MICDE)

Peer-Reviewed Publications

- **Nguyen, L.** & Ombadi, M. (2026) *Characterizing Patterns of Drought Synchronicity in the Contiguous United States*. Water Resources Research. https://doi.org/10.1029/2025WR041240
- **Nguyen, L.**, Todorovic, L., & Ombadi, M. (2026) *Anthropogenic Global Warming Increases the Risk of Record-breaking Extreme Precipitation Events in low-income Countries*. Nature Communications Earth & Environment. https://doi.org/10.1038/s43247-026-03649-y
- Ombadi, M., **Nguyen, L.**, Gaur, S., Gronewold, A., & Reich, P. (2026) *Increased Dependency on Extreme Precipitation in a Warmer Climate*. Water Resources Research. https://doi.org/10.1029/2025WR042627

Manuscripts in Preparation

- **Nguyen, L.** & Ombadi, M. *Intercomparison of Object-Oriented Drought Clustering Methods*. In preparation.
- **Nguyen, L.** & Ombadi, M. *Drought Synchronization in a Warming World and Impacts on Breadbasket Regions*. In preparation.

Research Experience

Hydrological Extremes with HEIDA Lab — University of Michigan
Advisor: Mohammed Ombadi 2024 - present

- **Global Drought Synchronization and Breadbasket Exposure** (ongoing): Investigated how warming changed the risk of synchronized droughts across major crop-producing regions, using CMIP6 projections. Investigated the relative roles of precipitation deficits and atmospheric evaporative demand in drought synchronicity by comparing SPI- and SPEI-based patterns.
- **Intercomparison of Drought Clustering Methods** (ongoing): Compared methods for identifying and tracking drought events in space and time; assessed how method parameterizations affected event duration, spatial extent, intensity, and migration to distinguish robust drought characteristics from methodological differences.
- **Drought Synchronicity over the Contiguous United States** (Apr-Dec 2025): Characterized the spatio-temporal synchronization of meteorological droughts over the contiguous United States. Examined links between drought synchronicity and large-scale climate variability using event coincidence analysis, wavelets, and Random Forest/SHAP.
- **Record-Breaking Extreme Precipitation Events** (Apr 2024-May 2025): Assessed how global warming changed record-breaking precipitation events from a hazard/exposure/vulnerability perspective, combining CMIP6 projections with socioeconomic data.

Urban Heat Adaptation with AGU: Thriving Earth Exchange — Boynton Beach, FL
Community Science Fellow 2025 - present

- **Hot Zones: Urban Heat Resilience and Cooling-Center Planning**: Led a project to create a detailed roadmap for implementing cooling centers for Boynton Beach. Oversaw a nine-person team to research the risk of urban heat and prioritize objectives between researchers, municipal leaders, and non-profits.

Terrestrial Response Method Comparison with IGCB — **University of Michigan**

School for Environment and Sustainability — 2025-present

- **Multi-modal Comparison of Global Change Parameters:** conducted a systematic literature review to intercompare experimental, observational, and modeled terrestrial ecosystem responses to global warming, precipitation, and elevated CO_2 to assess the consistency of findings across approaches.

SAI modeling with FORCEE Research Internship — **Sandia National Laboratories**

Supervisor: Lauren Wheeler — May 2023 - Oct 2023

- **Stratospheric Aerosol Injection Modeling**: Investigated temperature control through stratospheric aerosol injection by validating an improved E3SMv2 aerosol chemistry as well as implementing a proportional-integral feedback controller to adjust injection rates against three temperature targets.

Tropical Stratospheric Dynamics with ADMG — **University of Michigan**

Advisor: Christiane Jablonowski — May 2022 - Oct 2023

- **Stratospheric Circulation and Transport in DOE's E3SMv2**: Assessed how accurately the model represented tropical stratospheric circulation and transport using Transformed Eulerian Means to investigate the Brewer Dobson Cycle. Identified a dry bias and overly rapid upward propagation of water-vapor anomalies.

Power Spectra of Precipitation with Brian Arbic's Group — **University of Michigan**

Advisor: Brian Arbic — Aug 2021 - Jan 2022

- **Frequency Analysis of High-Intensity Precipitation over Land vs Ocean**: Evaluated how well high-resolution climate models captured rainfall intensity from minutes to decades by comparing precipitation power spectra with high-frequency NOAA rain-gauge observations over land versus ocean.

Posters/Presentations

- **Nguyen, L. et al.** (*AGU 2026*) *Bridging Earth Observation and Community Knowledge: A Heat Resilience Case Study in Boynton Beach, Florida*, American Geophysical Union (AGU) 2026, Abstract 2095314: GH001. *Accepted.*
- **Nguyen, L.** (*March 17, 2026*) Invited speaker for *Student Stories from COP30: COP30 Debrief with Environmental Policy Association*, University of Michigan Environmental Policy Association
- **Nguyen, L.** & Ombadi, M. (*January 29, 2026*), *Quantifying Global Synchronization of Droughts in Breadbasket Regions Under a Warming Climate*, American Meteorological Society (AMS) Annual Meeting, Abstract 469617
- **Nguyen, L.** & Ombadi, M. (*December 19, 2025*), *Characterizing Drought Synchronicity Across the Contiguous United States (1980–2021)*, American Geophysical Union (AGU) 2025, Abstract 1969758: NH51E-0228
- **Nguyen, L.**, Li, Q., & Ombadi, M. (*December 18, 2025*), *Decoding Drought Dynamics: A Multi-Method Intercomparison for Event Characterization Across Spatio-Temporal Scales* , American Geophysical Union (AGU) 2025, Abstract 1969758: H41M-1332
- Todorovic, L., **Nguyen, L.**, & Ombadi, M. (*December 17, 2025*), *Disproportionate Increase of Unprecedented Rainfall Events in Low Income Countries*, American Geophysical Union (AGU) 2025, Abstract A31E-2108
- Gaur, S., **Nguyen, L.**, Gronewold, A., Reich, P., & Ombadi, M. (*December 15, 2025*), *Changes in Extreme and Annual Precipitation under Global Warming*, American Geophysical Union (AGU) 2025, Abstract GC11G-0662
- Zhong, Y., Li, Q., **Nguyen, L.**, & Ombadi, M. (*December 18, 2025*), *Examining the Impact of Drought on Vegetation in the Contiguous United States*, American Geophysical Union (AGU) 2025, Abstract B41I-1970
- **Nguyen, L.** (*August 13, 2025*) Invited speaker for *Webinar: Science with Purpose: How Scientists Can Help Empower Community Solutions*, American Geophysical Union, Thriving Earth Exchange
- **Nguyen, L.**, Ombadi, M. (2024), *Quantifying Global Event Synchronization of Droughts in a Warming Climate*, American Geophysical Union (AGU) 2024, Abstract 1620436: GC33G-0234
- Jablonowski, C., Hollowed, J., **Nguyen, L.**, Hughes, O., Ehrmann, T., Wagman, B., & Hillman, B. (2023), *A Closer Look at E3SMs Stratosphere: Circulation Biases, Transport, Tracers*, DoE CLDERA All-Hands Meeting, Albuquerque, NM, USA, Oct 17-19, 2023
- Jablonowski, C., Hollowed, J., **Nguyen, L.**, Ehrmann, T., Wagman, B., & Hillman, B. (2023), *Tropical Stratosphere-Troposphere Interactions in Selected CMIP6 Model*s, Joint SPARC DynVar - SNAP Meeting, Munich, Germany, Oct. 9-13, 2023
- Jablonowski, C., **Nguyen, L.**, Kinnison, D., & the CLDERA project team (2023), *Characteristics of the stratospheric tropical circulation in DoEs Energy Exascale Earth System Model E3SMv2*, CESM Workshop 2023, Boulder, CO, USA, June 12-14, 2023

- Jablonowski, C., **Nguyen, L.**, Golaz, J.-C., Rosenbloom, N., & Meehl, G. A. (2023), *Characteristics of the Stratospheric Tropical Circulation of the Energy Exascale Earth System Model E3SMv2*, European Geosciences Union (EGU) 2023, Abstract EGU23-15173

Academic Service

- Peer Reviewed 2 articles for Water Resources Research

Community Leadership

AGU: Thriving Earth Exchange **Detroit, MI**
Volunteer Community Scientist Aug 2024 - Dec 2025
- **Illegal Dumping Mitigation and Environmental Education**: Supported waste mitigation efforts at Scott Memorial Church through research, recycling workshops, and coordination with local muralists.

Climate Blue at University of Michigan **Ann Arbor, MI**
COP30 Student Delegate May 2025 - present
- Represented Umich as a Climate Blue delegate to the UNFCCC COP 30, producing accessible briefings and outreach materials to connect global climate negotiations with students, researchers, and local communities.

University of Michigan: CLaSP GUSTO **Ann Arbor, MI**
DEI Officer and Volunteer Graduate Mentor Aug 2024 - present
- Organized DEI related activities and mentorship within the department.
- Volunteered to mentor graduate students interested as part of the GUSTO mentorship program.

Ypsilanti Math Corp at University of Michigan **Ann Arbor, MI**
Course Instructor Summer 2024
Math Corp Activities/HW Volunteer May 2019 - Dec 2024
- Developed and implemented "Ice Cream" class curriculum on graph theory for 8th graders and high school to teach fun topics in math during the summer camp.
- Co-organized the activities room to increase awareness and engagement with mathematics for Ypsilanti children grades 7-9 at the Ypsilanti Math Corp summer camp and Super Saturdays.

FATE Give Merit Program **Ann Arbor, MI**
Mentor Sep 2022 - Dec 2024
- Mentored high school students weekly in Detroit, working on strengthening their professional skills.

University of Michigan: Applied Physics Department **Ann Arbor, MI**
Volunteer Graduate Mentor Aug 2022 - present
- Mentored incoming graduate student, Shriya Sinha. Set up weekly HW sessions and monthly dinners.

Physics Diversity, Equity, and Inclusion Committee **Ann Arbor, MI**
Outreach Committee Volunteer Aug 2021 - present
- Volunteered for outreach opportunities with the local communities. (WISE middle school demos, etc)

Michigan Math Circle **Ann Arbor, MI**
Volunteer Assistant Aug 2022 - May 2023
- Guided elementary to high school students in engaging and complex math topics.
- Gave interactive lectures and activities on accessible, but interesting math problems

Applied Physics Student Council **Ann Arbor, MI**
Authorized Signer Aug 2022 - present
- Responsible for activities hosted by the Applied Physics Department's graduate student council.

University of Michigan: College of Engineering **Ann Arbor, MI**
Volunteer Graduate Mentor Aug 2021 - Aug 2022
- Volunteered to mentor undergraduate students interested in climate and space. Set up regular HW sessions and talks to discuss various aspects of academic and social life.

Cal Women's Rugby Team **Berkeley, CA**
Academic Officer/(Player Position: Prop) Aug 2018 - Jun 2019
- Monitored and created individual academic plans, with an emphasis on fostering time management skills. Scheduled and organized study sessions and events to promote an academic lifestyle.

Mathematics Undergraduate Student Associations (MUSA) **Berkeley, CA**
Berkeley Undergraduate Mathmentor Program (BUMP) Mentor Aug 2018 - Jun 2019
- Each semester, mentored two underclass students interested in mathematics in order to foster enthusiasm for continuing math education. Met twice a week to talk about school/hobbies/life.

Skills

- **Computing**: Python (xarray, NumPy, pandas, scikit-learn, PyTorch, Dask, Matplotlib), Bash, Git, LaTeX, HPC batch environments, and terabyte-scale NetCDF workflows.
- **Methods and models**: Event Coincidence Analysis, clustering, Random Forest/SHAP, wavelet and spectral analysis; E3SMv2, CESM, WRF; CMIP6, HighResMIP, reanalysis, and station observations.
- **Additional tools**: AWS, Power BI; working knowledge of MATLAB, Fortran, C++, R, and SQL.

Teaching

- **Physics 141: Elementary Lab** (GSI) - Winter 2024

Work Experience

RSAE Labs **Tampa Bay, FL**
Bookkeeper Jul 2021 - present

- In charge of payroll, accounts receivable, accounts payable, and financial reports. Streamlined several administrative processes, to facilitate transparency.

Omnium **San Francisco, CA**
Data Scientist Jul 2019 - Aug 2021

- Classification of SKU rationalization using various clustering models.
- ETL on Consumer Package Goods (CPG) consumption data to derive key insights for client business intelligence.
- Optimized pivotal client business metrics (both top and bottom line) given budget constraints by building a heuristic algorithm to a Mixed Integer Non-Linear programming problem.
- Restructured and reformatted internal data capture processes and automated lower level analysis by establishing a coherent data model and pipeline using AWS and Power BI.
- Significantly streamlined and implemented intern training to create a systematic transition from training to internal and client facing company projects. Served as a project mentor for 8 different interns.

Sandia National Laboratories **Livermore, CA**
Nuclear Radiation and Detection Intern (NSSC affiliate) Aug 2018 - Jun 2019

- Oversaw and implemented a methodical study of neutron production in high-Z materials (lead and tungsten) on a per-muon basis, with the goal of calculating the neutron multiplicity distribution for both spallation and capture channels. Optimized the data acquisition system, independently researched and ran experiments.

Miscillaneous

- I am PADI advanced scuba certified and always looking for partners to explore the waters with!
- I have my Accelerated Free Fall (AFF) license to skydive
- I am an avid baker, board gamer